

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **XCTest-based automated test suites** using **Swift** for core application features across multiple Apple platforms, increasing test coverage by **18%**.
- Implemented **LLM-powered test generation** capabilities, integrating **Python** scripts with natural language processing models to identify edge cases and generate novel test scenarios.
- Executed comprehensive automated testing cycles, analyzing **detailed performance metrics** and providing actionable insights to development teams that reduced critical bug rates by **14%**.
- Integrated automation frameworks with **CI/CD pipelines**, ensuring seamless continuous deployment and enabling faster feedback loops for release engineering.
- Developed intelligent **defect analysis systems** by fine-tuning **machine learning models** on historical bug data to predict defect likelihood and root causes.
- Collaborated with cross-functional teams to define and refine test strategies for new feature development, ensuring alignment with product roadmaps and quality standards.
- Containerized automation execution environments using **Docker**, facilitating consistent and scalable test runs on various cloud infrastructure.

SKILLS

Programming Languages – Swift, Python, Java, SQL, R

Testing Frameworks – XCTest, JUnit, Mockito

AI/ML – GenAI, LLMs, Machine Learning Model Training, Fine-tuning, scikit-learn, Pandas, NumPy

DevOps & Cloud – CI/CD, Jenkins, Docker, AWS (EC2, S3, Lambda, CloudWatch), Kubernetes

Tools & Platforms – Jira, Confluence, Git, Splunk, Dynatrace, Power BI, Tableau

PROJECTS

Swift Test Automation Framework for iOS Features

- Engineered a reusable **Swift** test automation framework using **XCTest** to streamline testing for new iOS application features.
- Integrated the framework with **Jenkins** for automated execution and reporting, reducing manual testing effort for regression suites by approximately **40%**.
- Developed a suite of end-to-end tests covering user authentication and core data workflows, achieving over **85%** pass rate on initial runs.

Intelligent Defect Prediction Model

- Developed a **Python**-based machine learning model to predict software defect likelihood using historical JIRA ticket data and code commit metrics.
- Trained a **gradient-boosted classifier** on a dataset of ~150,000 tickets, achieving an AUC of **0.82** for identifying high-risk defects.
- Integrated model output into a reporting dashboard, enabling engineering managers to prioritize code reviews and testing efforts more effectively.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)